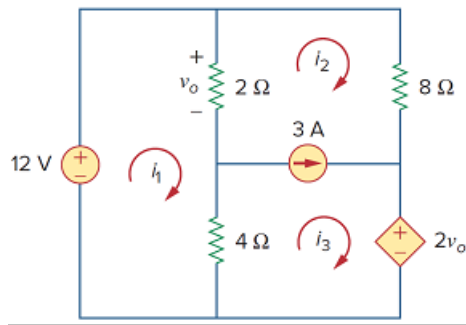


Problem

Use mesh analysis to find i_1 , i_2 , and i_3 in the circuit of Fig. 3.97.

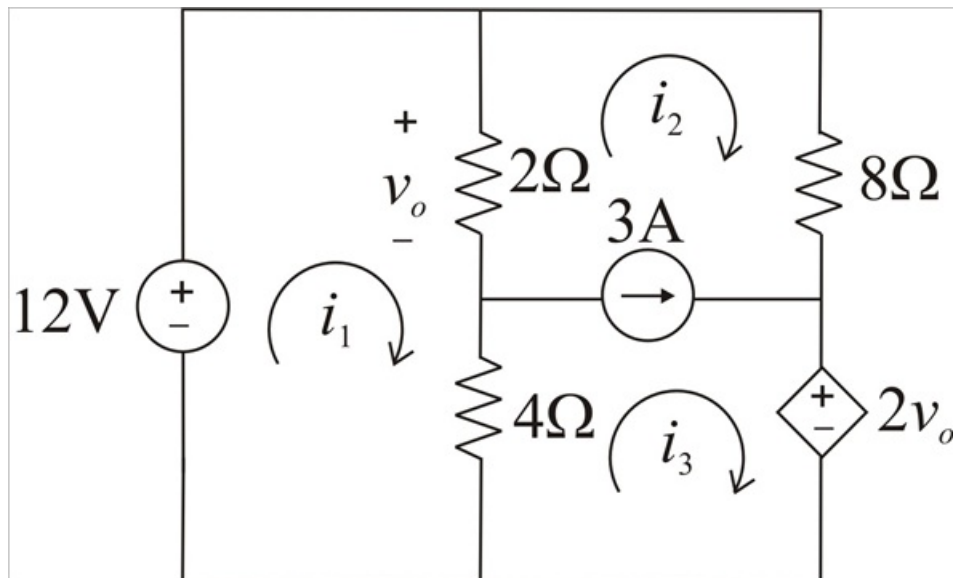
Figure 3.97:



Step-by-step solution

Step 1 of 7

Given circuit diagram is



Step 2 of 7

From independent current source we can write

$$i_3 - i_2 = 3 \quad \text{..... (1)}$$

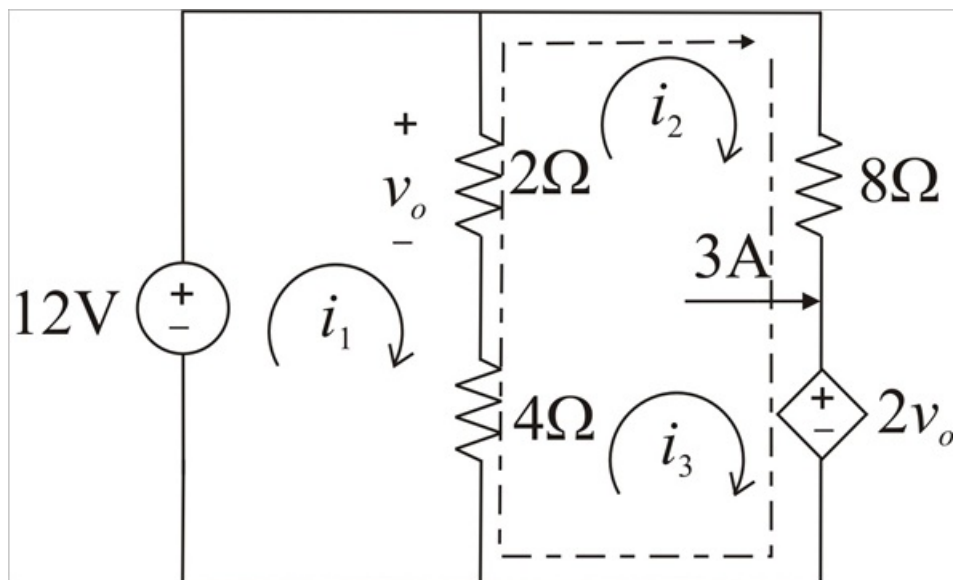
By using KVL to mesh 1 we can write

$$-12 + 2(i_1 - i_2) + 4(i_1 - i_3) = 0$$

$$6i_1 - 2i_2 - 4i_3 = 12 \quad \text{..... (2)}$$

Step 3 of 7

From the above circuit we can represent the super mesh as shown below



Step 4 of 7

By applying KVL to super mesh we can write

$$4(i_3 - i_1) + 2(i_2 - i_1) + 8(i_2) + 2v_o = 0$$

From the given circuit we can write

$$v_o = 2(i_1 - i_2)$$

Now above equation becomes

$$4(i_3 - i_1) + 2(i_2 - i_1) + 8(i_2) + 2(2(i_1 - i_2)) = 0$$

$$4i_3 - 4i_1 + 2i_2 - 2i_1 + 8i_2 + 4i_1 - 4i_2 = 0$$

$$-2i_1 + 6i_2 + 4i_3 = 0 \dots\dots\dots (3)$$

Step 5 of 7

From equation (1) we can write

$$i_3 = 3 + i_2$$

Now equation (2) becomes

$$6i_1 - 2i_2 - 4(3 + i_2) = 12$$

$$6i_1 - 6i_2 = 24$$

$$i_1 - i_2 = 4$$

$$i_2 = i_1 - 4$$

Step 6 of 7

Now equation (3) becomes

$$-2i_1 + 6(i_1 - 4) + 4(3 + i_2) = 0$$

$$-2i_1 + 6(i_1 - 4) + 4(3 + i_1 - 4) = 0$$

$$-2i_1 + 6i_1 - 24 + 12 + 4i_1 - 16 = 0$$

$$8i_1 = 28$$

$$\boxed{i_1 = 3.5\text{A}}$$

Step 7 of 7

$$i_2 = 3.5 - 4$$

$$\boxed{i_2 = -0.5\text{A}}$$

$$i_3 = 3 - 0.5$$

$$\boxed{i_3 = 2.5\text{A}}$$